Adaptive Schema Domains: Validation and Higher-Order Structure of the Young Positive Schema Questionnaire in a Sample of Flemish Emerging Adults

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Abstract

Central to schema theory (Young et al., 2003) is the idea that schemas can be grouped into domains. The organization of these schemas into domains has clinical importance, but debate persists in the literature. This study uses the Young Positive Schema Questionnaire (YPSQ; Louis et al., 2018) to explore the higher-order structure of adaptive schemas, paralleling the literature on maladaptive schema domains. After validating a Dutch version of the YPSQ in a large Flemish emerging adult sample, a four-domain model (Bach et al., 2017) was compared to a five-domain model (Young et al., 2003). Confirmatory factor analysis validated the 14 adaptive schemas of the YPSQ. Measurement invariance across gender and age groups was confirmed. Discriminant validity (through associations with maladaptive schemas and psychopathology), convergent validity (through associations with well-being measures), and incremental validity (explained variance above and beyond maladaptive schemas) were demonstrated. Both the hypothesized four-domain and five-domain models showed acceptable fit in structural equation modeling, with statistical comparisons favoring the slightly better fit of the four-domain model, as expected (Louis et al., 2020; 2024). Implications for schema theory and clinical practice are discussed.

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Introduction

Maladaptive and Adaptive Schemas

The term "schema" has a rich history. Developmental researchers as early as Piaget (1929) and Bartlett (1932) stated that past experiences help us make sense of new experiences by supplying us with expectations and "frameworks" for action. Aaron Beck (1967) applied the concept of schemas to gain a cognitive understanding of depression, arguing that schemas underly the negative interpretation and organization of information which results in negative views of the self, others or the world. Jeffrey Young (Young et al., 2003) expanded on these ideas and added a developmental dimension to schemas, as he proposed that people have several core emotional needs, which can get frustrated by environmental influences (e.g., abuse, neglect, bullying) in interaction with personal characteristics (e.g., temperament), which subsequently results in the development of an "early maladaptive schema" (EMS). Young and his colleagues (2003) proposed 18 EMS and an assessment tool, the Young Schema Questionnaire (YSQ-S3; Young & Brown, 2005), was developed and has been validated across countries and cultures (e.g., Baranoff et al., 2006; Cui et al., 2011; Lee et al., 1999; Soygüt et al., 2009).

Despite the great importance of investigating risk factors associated with psychopathology, such as EMS, researchers are increasingly interested in those factors that contribute to development going "right", sometimes despite strong risk factors (Masten & Barnes, 2018; Seligman & Csikszentmihalyi, 2000). To get a nuanced view on how psychopathology relates to well-being, it is necessary to look at both risk and protective factors. From a need-based perspective, this implies that we need to look beyond the effect of need frustration or the lack thereof, but explicitly consider the role of need satisfaction (Chen et al., 2015; Lockwood & Perris, 2012). Low need frustration may help explain why psychopathology does not manifest, but the effects of need satisfaction need to be considered to understand why people flourish. This also has clinical implications insofar that clinicians should not only correct need frustration, but also look at how they can foster need satisfaction.

From the view of schema theory, a well adjusted person is one whose core emotional needs were adequately in childhood met , leading to the formation of a balanced sense of self and the ability to engage positively with others, as well as the capability to continue meeting their adult needs effectively. In other words, such a person has adaptive schemas regarding themselves and their relations with others. Given the fact that core emotional needs can not only be frustrated but are often supported throughout childhood and adolescence, Louis et al. (2018) conceptualized “early adaptive schemas” (EAS) as positive counterparts for Youngs EMS. As such, the Young Positive Schema Questionnaire (YPSQ; Louis et al., 2018) was developed. First, a work group of researchers and schema therapists proposed 18 EAS, each representing an adaptive counterpart of a maladaptive schema. This resulted in an initial pool of 96 items. Using exploratory and confirmatory factor analyses on data from five international adult samples, a final pool of 56 items representing 14 EAS was obtained. This 14-factor model had satisfactory psychometric properties. Note that despite EAS and EMS being conceptual counterparts, their development and activation by triggers can occur independent from each other, as need frustration can happen parallel to need satisfaction. Indeed, Louis et al. (2018) found that EAS predicted measures of psychopathology and measures of well-being above and beyond age, gender and all EMS, providing validity for adaptive schemas as a separate construct and for the YPSQ as a valid questionnaire to measure them. The YPSQ showed adequate internal consistency, discriminant and construct validity, and its 14-factor structure has been replicated in additional international samples (Louis et al., 2020; 2023). However, three subsequent independent studies did not replicate this factor structure. Using exploratory analyses on the 56 items developed by Louis et al. (2018), Huckstepp et al. (2023), in an eating disordered German sample, found a nine-factor model, and Paetsch et al. (2022) found 10 factors in both a community and a psychiatric German adult. Moreover, in a Chinese study and using confirmatory analyses, Chi et al. (2022) identified an 11-factor model. As such, more research is needed to firmly establish the structure of the YPSQ. However, all three studies did conclude that EAS have the potential to be a valuable addition to psychopathological research and therapeutic practice.

Higher-Order Domains

An important assumption in Young's schema theory is that EMS are centered around a higher-order theme, each of which corresponds to the frustration of a core emotional need (Young et al., 2003). There has, however, been much debate in the literature on the amount and content of these domains. Using exploratory methods, different versions of the YSQ and different factor-analytic techniques have resulted in two (Saariaho et al, 2012), three (Calvete et al., 2013), four (Bach et al., 2017; Lee et al., 1999; Unoka et al., 2007; Yalcin et al., 2020) or five (Soygüt et al., 2009) domain solutions. However, the different usage of methodologies resulted in different interpretation and compositions of the domains, even if the same amount of domains is proposed. Young et al. (2003) initially proposed five schema domains based on core emotional needs, but support for this model has been weak. Only Young (1999) and Calvete et al. (2013) found a good fit, while other studies reported poor fit (e.g., Hawke et al., 2012; Kriston et al., 2012; Sakulriprasert et al., 2016). To address this, a revised four-domain model was suggested (Bach et al., 2017), which has received support from a meta-analysis of 27 studies by Thimm et al. (2022). However, the authors also called for further research and theoretical refinement.

The Present Study

In the development of the YPSQ, Louis et al. (2018) developed their 14 schemas based on the revised model (Bach et al., 2017), and proposes a four domain solution for EAS: *Connection/Acceptance, Healthy Autonomy, Reasonable Limits, Healthy & Realistic Standards.* This mirrors an earlier proposition on this issue by Bach et al. (2017) as well as Lockwood & Perris (2012). Using exploratory factor analysis on the 14 factors, Louis et al. (2020) found that a four-domain solution for the 14 EAS best fitted the data. Moreover, the authors argue that these domains mirror those proposed by Bach et al. (2017). Subsequently, the proposed four domain solution was confirmed in additional international samples and it demonstrated a better fit compared to a five domain model (Louis et al., 2020; 2024). However, after the exploratory analyses (Louis et al., 2020), 5 out of the 14 EAS did not share the same domain within the studies of Louis and colleagues and Bach and colleagues nor do they share the domain with their maladaptive counterpart, which somewhat undermines the call for theoretical refinement. As such, more research on EAS domains is warranted.

A case could be made to opt for a confirmatory approach to the question of the number and content of schema domains, as a clear framework for the amount of EAS (Louis et al., 2018) and their organization into domains (Bach et al., 2017) is available. Therefore, in this study, after establishing the factor validity of a Dutch translation of the YPSQ in a Flemish sample, it was hypothesized that a four-domain model would fit the data better compared to a five-domain model (Bach et al., 2017; Louis et al., 2020; 2024; Thimm, 2022). Measurement invariance for age groups and gender will also be assessed. To further assess the usefulness of the YPSQ and EAS as a construct, it was hypothesized that the discriminant and convergent validity of EAS as measured with the YPSQ would be demonstrated respectively through negative associations with measures of maladaptive schemas and psychopathology and through positive associations with measures of well-being. In line with the assumption that EAS and EMS are related but distinct constructs, it is expected that these negative correlations will be significant, but not too substantial (>.80; Louis et al., 2018). In addition, it was hypothesized that adaptive schemas will add explained variance in the prediction of psychopathology above and beyond age, gender and EMS (i.e. demonstrating incremental validity).

Previous research on the YPSQ (Huckstepp et al., 2023; Louis et al., 2018; Paetsch et al., 2022) used age ranges spanning the entire adulthood. However, late adolescence and early adulthood may be a particularly relevant age period for EAS, given that the transition to adulthood can be considered a period of "storm and stress" (Arnett, 2000) where events cause existing schemas to get triggered and reinforced (Young, 2003). Moreover, in accordance with Youngs schema theory, research shows that personality disorders (for which schema-based therapy was originally designed) often find their onset in adolescence and crystalize in emerging adulthood (e.g., Chanen & Thompson, 2019; Videler et al., 2019). This makes it important to have measurement tools that are adequate and adjusted for those populations. Therefore, the present study focuses on the distinct age group of late adolescence/emerging adulthood.

Method

Participants & Procedure

Throughout the summer of 2023 a community sample of 711 late adolescents and emerging adults was recruited. The sample included senior year secondary education students (47.1%; recruited through high schools), higher education students (44.9%; recruited through the institutions' communication channels) and non-studying/working emerging adults (8%; recruited online). The participants had to fill out an online test battery in Dutch consisting of questionnaires relating to EAS, EMS and psychosocial adjustment, as well as demographics, which took approximately 40 minutes. Instructed response items (e.g., "choose response option four") were added to assess whether participants filled out the questionnaires in a focused manner. In total, 1258 people responded to the questionnaire. Respondents were excluded if they did not completely fill out the YPSQ or did not correctly respond to two out of three instructed items. Finally, a sample of 711 youngsters remained, consisting of 68.9% females with an average age of 19.53 years. See Table 1 for an overview of the sample characteristics.

Parents of minor participants gave passive informed consent, and all participants gave active informed consent before filling out the questionnaires. Gift certificates were raffled to incentivize participation. This procedure was approved by the Ethical Commission of the authors' institution (Faculty of Psychological and Pedagogical Sciences, Ghent University, Belgium) [Approval Number: 2022-104].

Measures

Adaptive Schemas

A Dutch translation of the original YPSQ (Louis et al., 2017) was created according to the guidelines of the International Testing Commission using the Conference Method (Hambleton, 1994; see supplemental material for detailed overview of the translation process), and as such a 56-item 14-subscales self-report questionnaire was obtained. Permission to translate was granted by the original author. Example items include "For the most part, I have had someone who really listens to me, understands me, or is tuned into my true needs and feelings" (*emotional fulfillment*) and "I usually feel that I'm not in any danger and that things will be OK" (*basic health and safety/optimism*). A six-point Likert Scale ranging from 1 (*completely untrue of me*) to 6 (*describes me perfectly*) is used. Previous studies indicated good reliability and validity in adult community English-speaking samples (Louis et al., 2018; 2023; 2024).

Maladaptive Schemas

The Dutch version of the Young Schema Questionnaire – Version 3 Short Form (YSQ-S3; Young & Brown, 2005) consists of 18 subscales, representing the 18 EMS. A six-point

Likert Scale ranging from 1 (*completely untrue of me*) to 6 (*describes me perfectly*) is used. Example items include “No one I desire would want to stay close to me if he or she knew the real me” (*defectiveness/shame*) and “I think that if I do what I want, I’m only asking for trouble” (*subjugation)*. Previous studies have shown that the YSQ-S3 has good internal consistency and validity (e.g., Bach et al., 2015; Bosmans et al., 2010). In the present study, the internal consistency of all schemas ranged between .71 (*approval* seeking) and .89 (*social isolation*). However, the *entitlement* schema did not achieve sufficient internal consistency (.57). Therefore, this schema was left out of any further analyses.

Psychopathology

The extent to which participants experienced symptoms of *Aggression, Anxiety* and *Depression* was measured with the Dutch version of the Symptom Checklist 90 – Standard (SCL-90-S; Franke, 2014), a self-report questionnaire for psychological complaints in (young) adults. 34 items were answered on a five-point Likert scale (from zero = *“not at all”* to four = *“very much”*). The respective internal consistencies were $\alpha = .84$, $\alpha = .92$ and $\alpha = .91$. Sample items include: “In the past seven days, I quickly became irritable or agitated” (*Aggression*), “… I was feeling tense or restless” (*Anxiety*) and “… I was not interested in anything” (*Depression*), The questionnaire has good test-retest reliability, content validity, concurrent validity and convergent validity (Franke, 2014; Georg et al., 2021).

Well-being

Measures of positive adjustment included the Subjective Vitality Scale (SVS; Ryan & Frederick, 1978) and Satisfaction With Life Scale (SWLS; Pavot & Diener, 1993), respectively consisting of seven ($\alpha = .76$) and five items ($\alpha = .86$) which are answered on a five-point Likert scale (from one = “*totally do not agree with”* to five = “*totally agree with”*). Sample items include “I am satisfied with my life” (SWLS) and “I feel very energetic lately” (SVS). Previous studies showed good internal consistency and validity of both questionnaires (Vansteenkiste et al., 2006; Van Petegem et al., 2013; Schwartz et al., 2011)

Demographics

All participants reported on their age, gender ("male", "female", "other"), and secondary education level ("academic-oriented secondary education", "technical secondary education", "vocational secondary education", "other").

Plan of Analysis

The hypotheses and plan of analysis were preregistered before accessing any data (https://doi.org/10.17605/OSF.IO/UZCME). The data that support the findings of this study are possible from the corresponding author, J.C., upon reasonable request.

Factor Structure

To assess the 14-factor structure of the YPSQ, a confirmatory factor analysis (CFA) was conducted in R using the lavaan package (Rosseel, 2012). The adaptive schemas were modeled as latent variables on which the respective items of each schema loaded. Several fit indices were used to assess if the model fitted the data, according to the cut-offs set by Hu and Bentler (1999). For an excellent fit, the SRMR should be less than .08, the CFI should be greater than .95 and the RMSEA should be less than .07. It has been argued that ordinal variables with five or more categories can be treated as numerical variables so that Robust Maximum Likelihood would be suitable as estimator (Rhemtulla et al., 2012; Robizsch, 2020). However, model comparisons revealed that the model fit improved significantly when the items were considered as ordinal variables. As such, in line with the method of Louis et al. (2018), all items were set as ordinal variables and Diagonally weighted least squares (DWLS) was used as estimator. DWLS is also robust for the non-normality we encountered with most YPSQ items. Listwise deletion was used to handle missing data. To evaluate whether factor structure of the Dutch YPSQ was equal across gender and across age, measurement invariance was tested at three levels: configural, factor loading and intercept invariance. Model fit changes were examined, with a change of ≥ -.01 in CFI, supplemented by a change of ≥ .015 in RMSEA or a change of ≥ .03 (between configural and factor loading model) or ≥.01 (for testing intercept invariance) in SRMR indicating non-invariance (Chen, 2007).

Convergent, Discriminant and Incremental Validity

Bivariate correlations between EAS and EMS were calculated to assess discriminant validity of the YPSQ. Bivariate correlations between EAS, subjective vitality and life satisfaction were calculated to assess convergent validity of the YPSQ and are thus expected to be positive. To assess the incremental validity, hierarchical multiple regression analyses were performed with depression, anxiety, aggression, and vitality and life satisfaction as dependent variables. Independent variables in Step 1 were age and gender, all EMS in Step 1 and all adaptive schemas in Step 3. A minimal value of ΔR^2 = .0225 (or 2.25%) should be achieved from the second to the third step of a regression analysis (Hunsley & Meyer, 2003) to show that positive schemas would demonstrate sufficient incremental validity in predicting psychopathology and psychological well-being.

Higher-Order Domains

Bach et al. (2017) argued that it is important to consider which factor analytic methods to use to assess schema domains, since different methods have contributed to contradicting research results. It can be argued that confirmatory approaches have preference given that they are more top-down and theory-driven, which is appropriate since Bach et al. (2017) provide a clear theory, both on the number of domains as well as their content, to test. On the other hand, other researchers argue that exploratory methods have preference when it comes to personality-like variables (Hopwood & Donnelan, 2010). Because Bach et al. (2017) provide a theoretical rationale with clinical relevance for schema organization into either four (Bach et al., 2017) or five (Young et al., 2003) domains, a confirmatory approach was preferred in this study compared to an exploratory approach. The 14 EAS were modeled as latent factors in the measurement model (this being the CFA described above) and subsequently used in the structural model to assess the higher-order domain structure. The same cut-offs for the fit indices were used as described above. First, both a four-domain solution and a five-domain solution was fitted. Then, to determine which of these non-nested models fitted the data best, several fit indices were compared (Louis et al., 2024).

Results

Validation of the Young Positive Schema Questionnaire

Missing Data

Missing data (ranging from .3% for YPSQ item 46 and 10.3% SCL-90-S item 30) were missing completely at random based on Little's test (Little, 1988; $\chi 2$ = 697.40; df = 1634; p = 1.00).

Factor Structure

CFA was performed to validate the 14-factor structure of the YPSQ. Each EAS was modeled as a latent factor on which the respective items of each schema loaded. The model showed an acceptable fit (SRMR = .05; RMSEA = .05; CFI = .99). The factor loadings of the items on the latent factors were all moderate or high (ranging from .66 to .96), and are shown in Table 2, together with Cronbach's alpha for each subscale (ranging from .75 to .93). The model was subsequently tested for measurement invariance across gender (male, female, other). Due to too little participants (n = 7) indicating their gender as "other", gender measurement invariance was only tested across those indicating their gender as male or female. No significant changes (Chen, 2007) in fit indices could be observed at the configural, factor loading or intercept level. ($\Delta CFI_{Metric-Configural} = -.001$; $\Delta SRMR_{Metric-Configural} = .002$; $\Delta RMSEA_{Metric-Configural} = .003$; $\Delta CFI_{Scalar-Metric} = .001$; $\Delta SRMR_{Scalar-Metric} = -.002$; $\Delta RMSEA_{Scalar-Metric} = -.004$). As such, measurement invariance across gender was established. Moreover, the sample was also split into two based on age (with sample mean as the cutoff) as to assess whether measurement invariance could also be established for older versus younger emerging adults. Here, too, no significant changes (Chen, 2007) in fit indices could be observed at the configural, factor loading or intercept level ($\Delta CFI_{Metric-Configural} = -.001$; $\Delta SRMR_{Metric-Configural} = .002$; $\Delta RMSEA_{Metric-Configural} = .003$; $\Delta CFI_{Scalar-Metric} = .002$; $\Delta SRMR_{Scalar-Metric} = -.002$; $\Delta RMSEA_{Scalar-Metric} = -.005$).

[TABLE 1 & 2 NEAR HERE]

Convergent, Discriminant and Incremental Validity

Given that all missing data was MCAR, an expectation maximization algorithm was used to impute the missing data for the bivariate correlations and regression analyses. As can be seen in Table 3 and in line with the hypotheses, EAS and EMS correlated negatively with each other. Bivariate correlations between EAS, life satisfaction, subjective vitality, aggression symptoms, anxiety symptoms and depression symptoms were calculated (see Table 4). All associations were in the hypothesized direction, providing evidence for the discriminant and convergent validity of the Dutch YPSQ.

Next, to assess the incremental validity of EAS, hierarchical multiple regression analyses were conducted. EAS explained statistically significant additional variance in the prediction of aggression, anxiety, depression, subjective vitality and life satisfaction. But for anxiety and depression, this additionally explained variance did not reach .0225 (see Table 5).

[TABLE 3, 4 and 5 NEAR HERE]

Higher-Order Structure of the YPSQ: Adaptive Schema Domains

To assess the higher-order domain structure, a four-domain and a five-domain solution were modelled (see Table 6). The former was based on the proposition of Bach et al. (2017) the was based on the domains as described in Young's original theory (Young et al., 2003). Indices of both models were acceptable, however the measures for Young's five domain model were slightly less robust, leading to the conclusion that the four domain model was best supported by the data (see Table 7).

[TABLE 6 AND TABLE 7 NEAR HERE]

Discussion

Central to schema theory (Young et al., 2003) is the idea that core emotional needs give rise to EMS when they are frustrated throughout childhood and adolescence. In the literature, however, there has been much debate on the content of these emotional needs, as well as which schemas center around which needs (Thimm, 2022). The present study tackled this research question by looking at the higher-order organization of EAS, which have gained an increasing amount of attention in the empirical literature in the past decade (Louis et al., 2018; 2020; 2024). Using a large sample (N = 711) of Dutch-speaking emerging adults from Flanders, the purpose of this study was to assess the higher-order structure of EAS by comparing a four-domain with a five-domain model. This was done after validating the factor-structure of the YPSQ of 14 EAS that was suggested by the original authors (Louis et al., 2018).

Results through CFA confirmed the 14-factor structure within the Dutch YPSQ. Item loadings on each schema were good, as were the internal consistencies of all subscales as measured with Cronbach's Alpha. The confirmation of 14 EAS was supported by Faustino and Louis (2024) but stands in opposition with the studies of Paetsch et al. (2021) and Huckstepp et al. (2022) who found different factor solutions. Compared to the original version of the

YPSQ, several EAS were collapsed onto each other or simply removed due to low factor loadings. Huckstepp et al. (2022) argued these discrepancies are due to differences in study samples. However, differences in methodology should also be considered, as both studies used EFA on the original 56 items while we performed CFA. Given that Louis et al. (2018) already did the work of constructing a solid and theoretically meaningful measurement tool for EAS and replicated its structure in additional international samples (Louis et al., 2020; 2022; 2024), we thought it more appropriate to try to confirm his proposed 14-factor structure. Interestingly, our confirmation of 14 higher-order factors also contradicts the findings of a Chinese study using CFA which, after modifying their model, arrived at 10 factors (Chi et al., 2022). Although the authors provide no explanation as to why a reduced number of domains fitted best in their study, cultural differences may have played a role.

Furthermore, measurement invariance across age (younger and older emerging adults; with 19.53 years as the cut-off, i.e., the sample mean) and gender (male & female) was confirmed at three different levels (configural, loadings and intercept). Evidence for discriminant and convergent validity of the EAS as measured with the YPSQ was found through negative correlations with EMS, through negative correlations with symptoms of psychopathology (depression, aggression, anxiety) and through positive correlations with life satisfaction and subjective vitality. Associations with EMS were significant, but not too high (.80 or higher; Louis et al., 2018) to be considered two ends of one continuum. Hierarchical regression analyses demonstrated incremental validity. This finding aligns with previous studies (Chi et al., 2022; Huckstepp et al., 2023; Louis et al., 2018, 2024; Paetsch et al., 2023). EAS explained a significant additional portion of variance ($\Delta R^2 > .0225$) in predicting aggression, life satisfaction, and subjective vitality, beyond the effects of age, gender, and EMS. The effects on depression and anxiety were statistically significant as well, but the additionally explained variance did not reach .0225. This was somewhat unexpected and is not in line with the findings of Louis et al. (2018). All in all, the current findings provide evidence for EAS as a valid construct, distinct from EMS, and for the YPSQ as a valid and reliable measurement tool in Flemish samples.

The main aim of this study was to validate a higher-order domain structure in which EAS could be organized. Parallel to how the EMS are grouped into four or five domains, previous research have sought to do the same with EAS, arriving at four domains as the best solution compared to five domains (Louis et al., 2020; 2024). However, more research is needed on the content of these domains and to further compare a four domain with a five domain solution. Therefore, the present study grouped the 14 EAS of the YPSQ into four and five adaptive domains, based on the proposed model by Bach et al. (2017). These domains represent core emotional needs that are systematically supported (in contrast with the maladaptive domains, which represent the frustration of core emotional needs). Results through CFA indicated that both the four and the five-domain solution fitted the data adequately. Further comparison based on information indices revealed that the four-domain solution fitted the data slightly better. This is in line with our hypothesis and corresponds with the literature on maladaptive schema domains which have generally favored four domains (Louis et al., 2024; Thimm, 2022).

Nonetheless, based on model fit alone, five adaptive schema domains also appear to be acceptable. Further theoretical refinement, combined with research that carefully considers which factor analytical technique should be used to test the theory, are still necessary. Interestingly, a new revision of schema therapy is underway by an international work group of researchers and schema therapists (Arntz et al., 2021). Herein, two core emotional needs are added, namely the need for fairness and the need for self-coherence, and subsequent new (mal)adaptive schemas. This revised schema theory is yet to be empirically tested. Nonetheless, one of the points Arntz and his colleagues make is that any organization of schemas into domains should be grounded in theory, as empirically derived solutions may be less meaningful to clinical practice. This is an argument for confirmatory approaches, rather than exploratory. Exploratory methods to assess schema organization should not be shunned in schema theory research, however the results should be interpreted insofar that they have ecological validity and further the theoretical framework which underpins schema therapy.

Notwithstanding the fact that the five domain model could not be rejected, this study seems to add to the growing consensus that four and not five core emotional needs should underpin schema theory (Thimm et al., 2022). This growing consensus is important, given that many clinicians still base their work on the original model, which may be suboptimal. Moreover, this study adds to the validity of EAS as a construct and the YPSQ as a measurement tool, strengthening the assessment and of positive psychological components of schema therapy (e.g., the "healthy adult" mode). The incorporation of positive schemas in schema therapy allows for a more nuanced understanding of mental (ill-)health and provides both clinicians and clients with clearer therapy goals, i.e. strengthening positive schemas rather than solely diminishing.

Limitations

This study has several limitations. First, the method for translating the YPSQ into Dutch (see the supplemental material for a detailed overview of the process) did not include back-translating the items. Second, the present study used a rather limited age range and had a majority of female participants. The narrow age range can be viewed as a strength, given the particular relevance of our sample's age group for the development of personality-related pathology, for which schemas are important underlying mechanisms. However, the age range, as well as the overrepresentation of women, may also undermine the generalizability of the results to other age groups. As such, future research should try this in a larger and more diverse sample across the lifespan. Another possible limitation of this study is the usability of this questionnaire in the Netherlands, as Dutch spoken there differs from Dutch spoken in Belgium regarding vocabulary, pronunciation, and some grammatical nuances. Translating into Dutch should be done with careful consideration of the target audience, as word choice, tone, and cultural context may vary significantly between the two regions, especially when it pertains emotionally loaded statements, as is the case with schemas. therefore a separate validation study of this translated YPSQ in the Netherlands is advocated. Moreover, the present study only used cross-sectional data. It would be interesting to see what the longitudinal predictive power of EAS are for psychological well- and ill-being above and beyond EMS. Finally, the EMS *Entitlement* was not included in any analyses due to inadequate internal consistency.

Conclusion

This study had the aim assessing the higher order structure of adaptive schemas by comparing a four versus a five domain model. This was done using the YPSQ in a sample of Dutch-speaking Flemish emerging adults. The results slightly favored the four domain solution. This study stressed the importance of method choice when assessing the higher-order structure of schemas and argues that the study of schema domains should be grounded in theory. Moreover, notwithstanding some limitations, this study provides further support for EAS as a valid construct separate from EMS and the YPSQ as a useful measurement tool in late adolescents and emerging adults.

The authors report there are no competing interests to declare.

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Table 1

Sample characteristics

Secondary Education level	High	57.9%	
	Medium	17,4%	
	Low	22.8%	
	Other	1,7%	
Gender	Male	30.1%	
	Female	68.9%	
	Other	1.0%	
Age		*M* = 19.53 (range = 16.38-25.90)	*SD* = 1.88
Depression		$\bar{x}$ = 1.11	*SD* = .83
Anxiety		$\bar{x}$ = .78	*SD* = .71
Aggression		$\bar{x}$ = .89	*SD* = .67
Life satisfaction		$\bar{x}$ = 3.23	*SD* = ..89
Subjective vitality		$\bar{x}$ = 2.82	*SD* = .70

Note. Low = vocational educational track, medium = technical educational track, high = academic educational track, other = e.g., art-focused education

Table 2

Standardized factor loadings of the YPSQ items on each latent variable (i.e. positive schemas) and the internal consistency of each subscale.

Schema	Item	Standardized Factor Loading on Subscale	Subscale Cronbach's Alpha
Emotional Fulfillment	1. For the most part, I have had someone who really listens to me, understands me, or is tuned into my true needs and feelings.	.75	.82
	2. For much of my life, I have felt that I am special to someone.	.74	
	3. In general, people have been there to give me warmth, holding, and affection.	.74	
	4. I'm confident that there is a man/woman I desire who would continue to love me, even if he/she saw my weaknesses.	.66	
	5. I know I can depend on the people closest to me to always be there for me.	.82	
Succes	6. I'm as intelligent as most people when it comes to work (or school).	.82	.88
	7. I'm as talented as most people are at their work.	.90	
	8. I am as capable as most other people in areas of work and achievement.	.91	
	9. When it comes to work (or school), I usually do as well as, or better than, other people.	.70	
	10. When it comes to achievement, I consider myself a competent person.	.78	
Empathic Consideration	11. When I have to go along with what others decide and can't do what I want, I can accept it without continuing to try to get my way.	.68	.79
	12. I am usually OK with not getting my way in a group decision.	.82	
	13. When I ask someone for something and the answer is "no," I'm usually comfortable accepting it without pushing to get my own way.	.74	
	14. I respect others wishes even when they are different from mine.	.81	
Basic Health & Safety/Optimism	15. I generally feel safe and secure – that nothing bad is going to happen to me (such as serious financial problems, illnesses, strangers hurting me, or catastrophic events).	.79	.86
	16. I usually feel that I'm not in any danger and that things will be OK.	.84	
	17. I feel confident that I will have enough money to get by in the future and don't worry about losing everything.	.71	
	18. There's no need to worry all the time; things generally work out pretty well.	.88	

	19. I'm usually relaxed about making decisions; I don't worry that something terrible will happen if I'm wrong.	.79	
Emotional Openness	20. When it comes to showing my emotions, the people I care about see me as capable of being expressive and spontaneous.	.80	.84
	21. The people who matter to me see me as capable of being open and comfortable showing my emotions.	.84	
	22. I'm usually comfortable expressing my feelings to others when I want to.	.82	
	23. I'm usually comfortable showing my positive feelings to others (e.g., physical affection, telling people I care about them) when I want to.	.78	
Self-Compassion	24. If I make a mistake, I can usually forgive myself; I don't feel that I deserve to be punished.	.90	.85
	25. When I make mistakes, I usually go easy on myself and try to give myself the benefit of the doubt.	.82	
	26. Even when I fail at something, I don't feel that I should be made to suffer for it.	.81	
Healthy Boundaries/Developed Self	27. I have been able to establish a life of my own, and am not overly involved with my parent(s) and their problems.	.74	.71
	28. I don't feel that my parent(s) are trying to live through me – they let me have a life of my own.	.82	
	29. I have been able to separate from my parent(s) and become an independent person, as much as most other people my age.	.74	
Social Belonging	30. I usually feel included in groups.	.88	.93
	31. I usually fit in with others.	.85	
	32. I feel as much a part of groups as I want to be.	.87	
	33. I generally feel as accepted by others as I want to be when I am around other people.	.92	
	34. I feel as connected as I want to be with other people.	.90	
Healthy Self-Control	35. I usually stick to my resolutions.	.82	.80
	36. I'm usually able to discipline myself to complete routine or boring tasks.	.69	
	37. If I can't reach a goal, I'm usually persistent and don't easily give up.	.76	
	38. I'm usually able to sacrifice immediate gratification or pleasure in order to achieve a long-range goal.	.69	
Realistic Expectations	39. I'm usually realistic when it comes to expectations for myself; I don't have to be among the best to be satisfied with what I've done.	.78	.87
	40. I like to do well but don't have to be the best.	.78	

	41. I don't have to be perfect; I can usually accept "good enough".	.80	
	42. I have realistic expectations of myself and usually feel OK about how I am doing.	.96	
Self-Directedness	43. What I think of myself matters more to me than what others think of me.	.79	.86
	44. I don't need a lot of praise or compliments from others to feel that I'm a worthwhile person.	.75	
	45. I am more focused on doing what matters most than getting people to think well of me.	.80	
	46. I value my own accomplishments even when other people don't notice them.	.96	
Healthy Self-Interest	47. I work hard and also leave time for relaxation and fun.	.79	.75
	48. While I enjoy doing things for the people I care about, I make sure I have time for myself too.	.75	
	49. I can be a good person and, at the same time, consider my own needs to be as important as those of others.	.85	
Stable Attachment	50. I feel confident that the people I'm close to won't leave or abandon me.	.91	.89
	51. I trust that people won't leave me, so I don't act needy and drive them away.	.89	
	52. I don't cling to the people I'm close to because I'm confident that they won't leave me.	.83	
	53. I am confident that most people I know will be loyal and not betray me.	.80	
Competence	54. I think of myself as an independent, self-reliant person, when it comes to everyday functioning.	.85	.87
	55. I feel confident about my ability to solve most everyday problems that come up.	.91	
	56. I feel capable of getting by on my own in everyday life.	.85	

Table 3

Bivariate correlations between adaptive and maladaptive schemas.

	Abandonment	Mistrust	Defectiveness /Shame	Emotional Deprivaton	Social Isolationi	Dependence	Vul-nerability	Em-neshment	Failure	Impaired Self-Control	Subjugation	Self-Sacrifice	Approval-Seeking	Pessimism	Emotional Inhibition	Unrelenting Stndards	Punitiveness
Emotional Fulfillment	-.44***	-.55***	-.64***	-.71***	-.56***	-.35***	-.32***	-.24***	-.41***	-.26***	-.51***	-.22***	-.11***	-.46***	-.53***	-.26***	-.42***
Succes	-.25***	-.21***	-.33***	-.23***	-.26***	-.35***	-.16***	-.18***	-.64***	-.28***	-.31***	-.14***	.00	-.26***	-.26***	-.10**	-.25***
Empathic Consideration	-.16***	-.14***	-.08*	-.02	-.08*	-.10**	-.13***	-.12**	.01	-.16***	-.03	.14***	-.24***	-.11**	-.03	-.11**	-.03
Basic Health & Safety/Optimism	-.50***	-.46***	-.45***	-.42***	-.42***	-.38***	-.59***	-.30***	-.37***	-.27***	-.42***	-.28***	-.19***	-.64***	-.29***	-.32***	-.35***
Emotional Openness	-.17***	-.31***	-.41***	-.41***	-.32***	-.22***	-.17***	-.08*	-.27***	-.18***	-.29***	.03	-.04	-.26***	-.70***	-.16***	-.25***
Self-Compassion	-.38***	-.32***	-.40***	-.32***	-.32***	-.31***	-.24***	-.18***	-.31***	-.20***	-.36***	-.24***	-.17***	-.41***	-.24***	-.45***	-.55***
Healthy Boundaries/Developed Self	-.32***	-.28***	-.28***	-.33***	-.28***	-.26***	-.28***	-.56***	-.27***	-.24***	-.41***	-.20***	-.12**	-.33***	-.25***	-.09*	-.18***
Social Belonging	-.26***	-.37***	-.47***	-.48***	-.43***	-.27***	-.22***	-.14***	-.32***	-.21***	-.36***	-.01	-.08*	-.33***	-.70***	-020***	-.29***
Healthy Self-Control	-.17***	-.09*	-.22***	-.12**	-.18***	-.23***	-.18***	-.13***	-.27***	-.64***	-.24***	.04	-0.13***	-.21***	-.18***	.11***	-.08*
Realistic Expecations	-.30***	-.28***	-.38***	-.28***	-.31***	-.27***	-.22***	-.16***	-.23***	-.15***	-.35***	-.18***	-.21***	-.32***	-.23***	-.60***	-.46***
Self-Directedness	-.49***	-.30***	-.40***	-.25***	-.28***	-.37***	-.29***	-.25***	-.32***	-.24***	-.45***	-.29***	-.37***	-.37***	-.21***	-.23***	-.28***
Healthy Self-Interest	-.40***	-.34***	-.44***	-.33***	-.33***	-.34***	-.23***	-.23***	-.36***	-.20***	-.46***	-.41***	-.12**	-.37***	-.33***	-.27***	-.35***
Stable Attachment	-.69***	-.63***	-.56***	-.51***	-.50***	-.35***	-.40***	-.27***	-.30***	-.22***	-.48***	-.33***	-.26***	-.52***	-.36***	-.30***	-.37***
Competence	-.37***	-.24***	-.38***	-.19***	-.31***	-.60***	-.32***	-.36***	-.35***	-.23***	-.40***	-.06	-.12***	-.33***	-.25***	-.05	-.25***

Note. The maladaptive schema ‘entitlement’ was left out of the analyses due to low internal consistency. P < .05*; p < .01**; p < .001***

Table 4

Bivariate correlations between adaptive schemas and measures of well- and ill-being.

	Agression Symptoms	Anxiety Symptoms	Depressive Symptoms	Subjective Vitality	Life Satisfaction
Emotional Fulfillment	-.38***	-.39***	-.54***	.40***	.53***
Succes	-.17***	-.23***	-.31***	.27***	.37***
Empathic Consideration	-.30***	-.12**	-.14***	.17***	.12**
Basic Health & Safety/Optimism	-.38***	-.54***	-.55***	.43***	.48***
Emotional Openness	-.21***	-.17***	-.29***	.34***	.35***
Self-Compassion	-.28***	-.33***	-.44***	.29***	.37***
Healthy Boundaries/Developed Self	-.29***	-.26***	-.34***	.26***	.31***
Social Belonging	-.24***	-.24***	-.36***	.39***	.41***
Healthy Self-Control	-.18***	-.17***	-.26***	.31***	.30***
Realistic Expecations	-.28***	-.32***	-.39***	.30***	.34***
Self-Directedness	-.31***	-.40***	-.44***	.33***	.37***
Healthy Self-Interest	-.30***	-.36***	-.47***	.35***	.41***
Stable Attachment	-.43***	-.47***	-.54***	.35***	.45***
Competence	-.22***	-.30***	-.33***	.27***	.32***

Note. P < .05*; p < .01**; p < .001***

Table 5

Hierarchical Regression Analysis of the Dutch YPSQ Predicting aggression, anxiety, depression, life satisfaction and subjective vitality.

	R^2	ΔR^2	ΔF
Agression Symptoms			
Step 1: age, gender	.04	.04***	14.25
Step 2: all maladaptive schemas	.35	.33***	21.23
Step 3: all adaptive schemas	.39	.05***	4.31
Anxiety Symptoms			
Step 1: age, gender	.07	.08***	29.11
Step 2: all maladaptive schemas	.49	.43***	35.08
Step 3: all adaptive schemas	.50	.02*	1.94
Depression Symptoms			
Step 1: age, gender	.04	.04***	13.82
Step 2: all maladaptive schemas	.53	.56***	56.85
Step 3: all adaptive schemas	.52	.02***	2.71
Life Satisfaction			
Step 1: age, gender	.06	.06***	21.51
Step 2: all maladaptive schemas	.41	.36***	25.59
Step 3: all adaptive schemas	.50	.08***	8.10
Subjective Vitality			
Step 1: age, gender	.07	.07***	26.62
Step 2: all maladaptive schemas	.34	.29***	18.10
Step 3: all adaptive schemas	.42	.09***	7.46

Note. The maladaptive schema ‘entitlement’ was left out of the analyses due to low internal consistency. $p < .05$*; $p < .01$**; $p < .001$***

Table 6

Schema domains and standardized factor loadings of schemas on domains.

4 domain model (Bach et al., 2017)	Conncection/Acceptance		Healthy Autonomy		Reasonable Limits		Realistic Standards			
	Emotional Fulfillment	.85	Basic Health & Safety/Optimism	.73	Empathic Consideration	.38	Realistic Expectations	.73		
	Social Belonging	.78	Competence	.63	Healthy Self-Control	.52	Self-Compassion	.75		
	Emotional Openness	.59	Healthy Boundaries/Developed Self	.55	Self-Directedness	.80	Healthy Self-Interest	.85		
			Succes	.53						
			Stable Attachment	.77						
5 domain model (Young et al., 2003)	Conncection/Acceptance		Healthy Autonomy		Reasonable Limits		Self-Directedness		Spontaneity	
	Emotional Fulfillment	.83	Basic Health & Safety/Optimism	.76	Empathic Consideration	.80	Self-Directedness	.80	Realistic Expectations	.72
	Social Belonging	.76	Competence	.65	Healthy Self-Control	.64	Healthy Self-Interest	.83	Self-Compassion	.74
	Stable Attachment	.83	Healthy Boundaries/Developed Self	.56					Emotional Openness	.59
			Succes	.54						

Table 7

CFA results for the four and five domain solutions.

	X^2	*df*	*p*	X^2/df	CFI	TLI	SRMR	RMSEA
4 domains	5726.33	1464	<.001	3.91	.982	.981	.059	.064
5 domains	5858.17	1460	<.001	4.01	.981	.980	.060	.066